

Predictive Insight Engd

1. What are supervised learning algorithms?
→ Algorithms trained on labeled data, where each input has a known correct output. The model learns the input-output mapping during training and uses it to predict outputs for new, unseen inputs.
2. Difference between Regression Algorithms vs Classification Algorithms
→ Regression predicts a continuous numeric value (e.g. house price). Classification predicts a discrete category or class label (e.g. Spam vs not spam).
3. Explain Simple Linear Regression
→ A method that models the relationship between one independent variable (x) and one dependent variable (y) as a straight line: $y = mx + c$. The line is fit by finding the slope (m) and intercept (c) that minimize the error between predicted and actual values.
4. List and Explain Assumptions of Linear Regression
→ Linearity - relationship between x and y is linear.
Independence - residuals (errors) are independent of each other.
Homoscedasticity - Residuals have constant variance across all values of x .
Normality - Residuals are approximately normally distributed.
No multicollinearity - Independent variables are not highly correlated with each other.

5. What is Bias-Variance Trade off?

→ Bias is the error from overly simple assumptions which cause underfitting. Variance is the error from a model being too sensitive to training data which causes overfitting.

6. Explain overfitting and underfitting with examples.

→ Overfitting - models learn the training data too closely, including its noise, and performs poorly on new data. Underfitting - model is too simple to capture real patterns.